



HAL 2021 Revision Marathon | Digital Electronics | Part 1 | Gradeup



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Digital Electronics

Part -1



ABOUT ME : CHANDAN JHA

- **10+ Years' Teaching Experience**
- **Cracked ISRO Scientist/Engineer**
- **M.Tech, Communication & Networks**
- **Conducted 200+ Motivational Session including NIT's and IIT's**
- **Area of Expertise: Digital Circuits, Control Systems, Signals and Systems, Communication, Microprocessor**



10+ Gradians
Under **AIR 10**



Syllabus

1. Logic GATE

NOT, AND, OR, NAND, NOR, EX-OR, EX-NOR, Inhibitor, Inhibition.

2. Minimization → Boolean algebra

K-map

3. Combinational circuit

Comparator, MUX, De-MUX, Encoder, Decoder, half adder, full adder, half subtractor, full subtractor, serial adder, Parallel adder, look-ahead carry adder, Multiplier.

4. Sequential circuit

Latches, FF, Triggering of FF's, Design of FF's, Registers, Counters.

5. Analog to digital and digital to analog converter.

6. Number system

7. Finite state machine

8. Memory

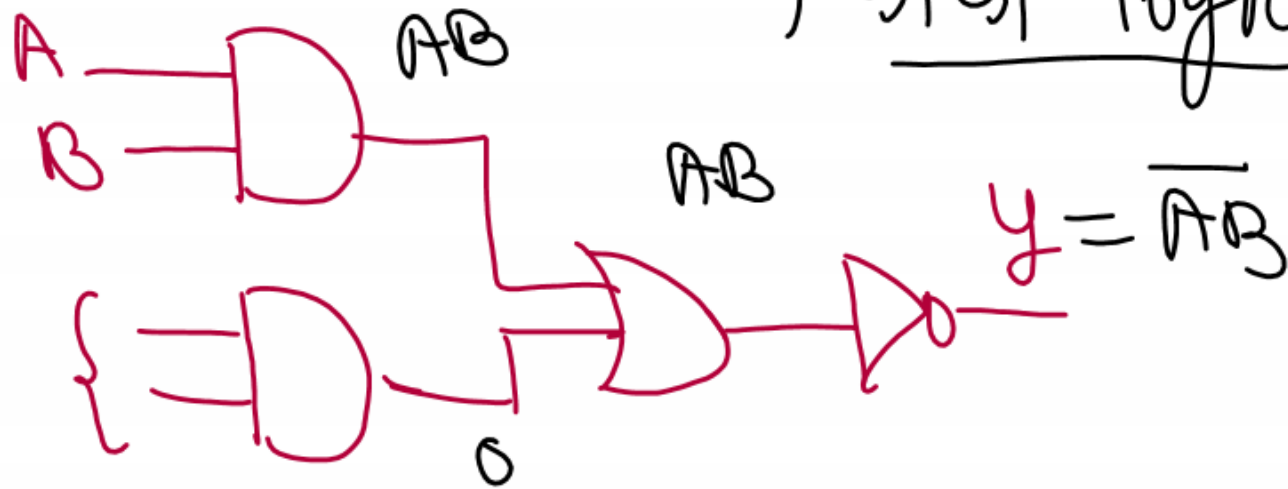
9. Logic Family

Logic GATE

TTL $\rightarrow$ floating terminal = high

ECL $\rightarrow$ floating terminal always acts = Low
fastest logic

ECL



① HTL (High threshold logic)

↳ Highest Noise Margin

② ECL (Emitter coupled logic)

↳ fastest logic

↳ Because Transistor Never goes into saturation.

(Non saturated logic)

③ I^2L (Integrated injected logic)

↳ Highest Packing density

↳ Highest FOM (figure of Merit)

as low as possible

for a good ckt.

$$\text{FOM} = \text{Propagation delay} \times \text{Power dissipation}$$
$$p_p = ns \times mJ/s$$

CMOS.

↳ ① Highest Fanout

② Lowest power dissipation

Logic gates.

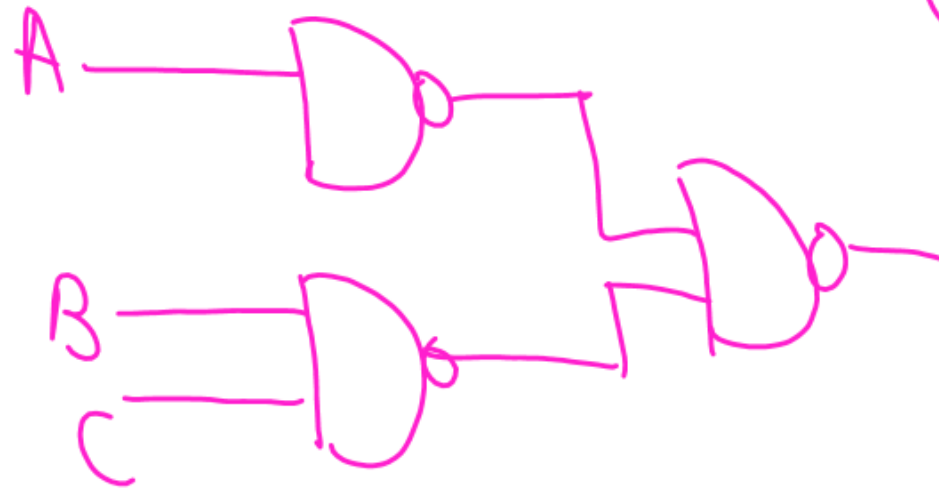
NAND
NOR

→ Minimum no. of NAND/NOR ✓

Q

$$f = A + BC$$

$$f = A \cdot A + BC$$



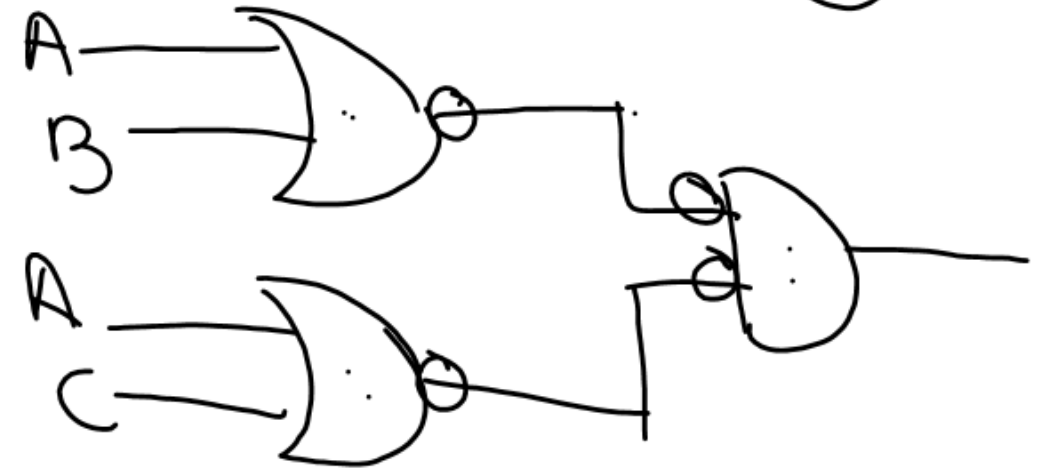
NAND = 3

NOR = 3

$$AB + CD$$

$$f = A + BC$$

$$f = (A + B) \cdot (A + C)$$



X-OR

$$\bar{A}B + A\bar{B}$$

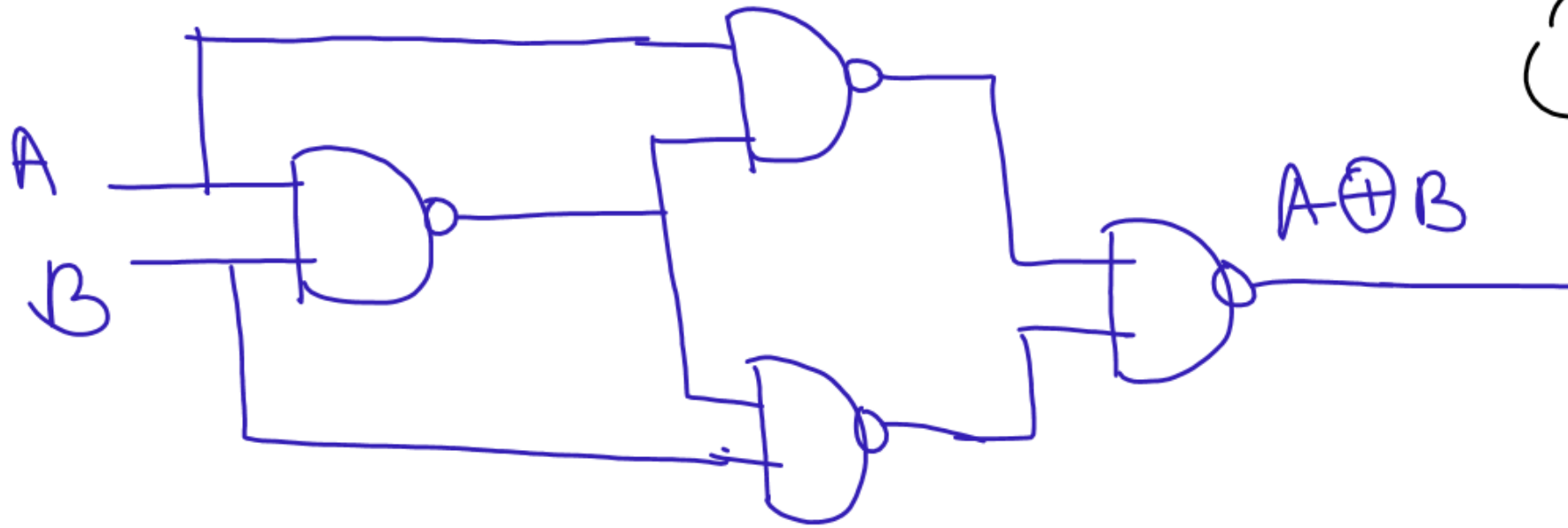
$(A+B)(\bar{A}+\bar{B}) \rightarrow \text{XOR}$

NAND $\rightarrow$ ④

NOR $\rightarrow$ ⑤

7:00PM - 10PM

④



$(A+B)(\bar{C}+\bar{D})$

④ $\rightarrow$ NAND

Q:

$$X \rightarrow (3)$$
$$(AB + CD) + EF$$

NAND 2

$$X \rightarrow (3)$$

(A) 4

~~(B) 6~~

(C) 8

(D) 10

$$X + EF$$
$$X \cdot X + EF \rightarrow (3)$$
$$\underline{\underline{6}}$$

X-OR

commutative ✓

associative ✓

$$\begin{aligned} A \oplus B &= \bar{A}B + A\bar{B} \\ &= (A+B)(\bar{A}+\bar{B}) \end{aligned}$$

$$\left\{ \begin{array}{l} A \oplus A = 0 \\ A \oplus 0 = A \\ A \oplus \bar{A} = 1 \\ A \oplus 1 = \bar{A} \end{array} \right.$$

Q If $A \oplus B = C$

$$A \oplus C = B$$

Then $A \oplus B \oplus C = ?$

$$A \oplus B \oplus C = \text{😊}$$

★ $A \oplus B \oplus C = \sum m(1, 2, 4, 7)$

↳ No further minimization

★ X-NOR

 $A \oplus B = \overline{A \odot B}$





$$A \odot B = \overline{A \oplus B}$$

$$\overline{A} \odot B = A \odot \overline{B} = \overline{A} \oplus \overline{B} = A \oplus B$$

$$\overline{A} \oplus B = A \oplus \overline{B} = \overline{A} \odot \overline{B} = A \odot B$$

✓✓ X-NOR

$$A \odot B = \bar{A}\bar{B} + AB$$
$$= (A + \bar{B})(\bar{A} + B)$$

{ Even input X-NOR

$$A \odot B = \overline{A \oplus B}$$

$$A \odot B \odot C \odot D = \overline{A \oplus B \oplus C \oplus D}$$

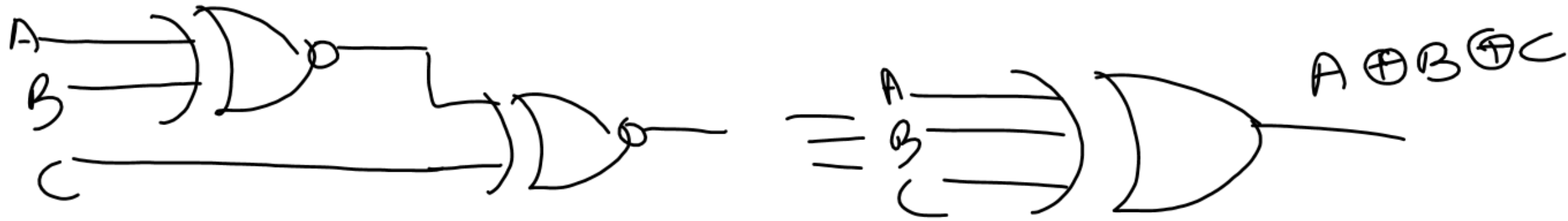
Even input X-NOR

When even no. of
input high = o/p high

Odd no. of input

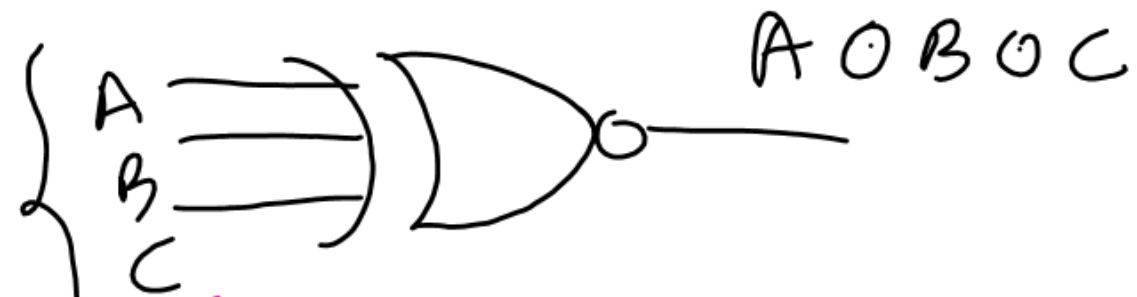
high = o/p Low

✓ Odd no. of input



$$\underline{(A \oplus B) \oplus C} = A \oplus B \oplus C$$

→ odd no. of 1 in input → o/p high
even no. of 1 in input → o/p low



$$A \oplus B \oplus C = \sum m(1, 2, 4, 7)$$

X $(A \oplus B \oplus C = \sum m(0, 3, 5, 6))$

→ even no. of 1 in input - output high
 Odd no of 1 in input - o/p low

$$A \oplus B = X$$

$$X \odot \bar{X} = 0$$

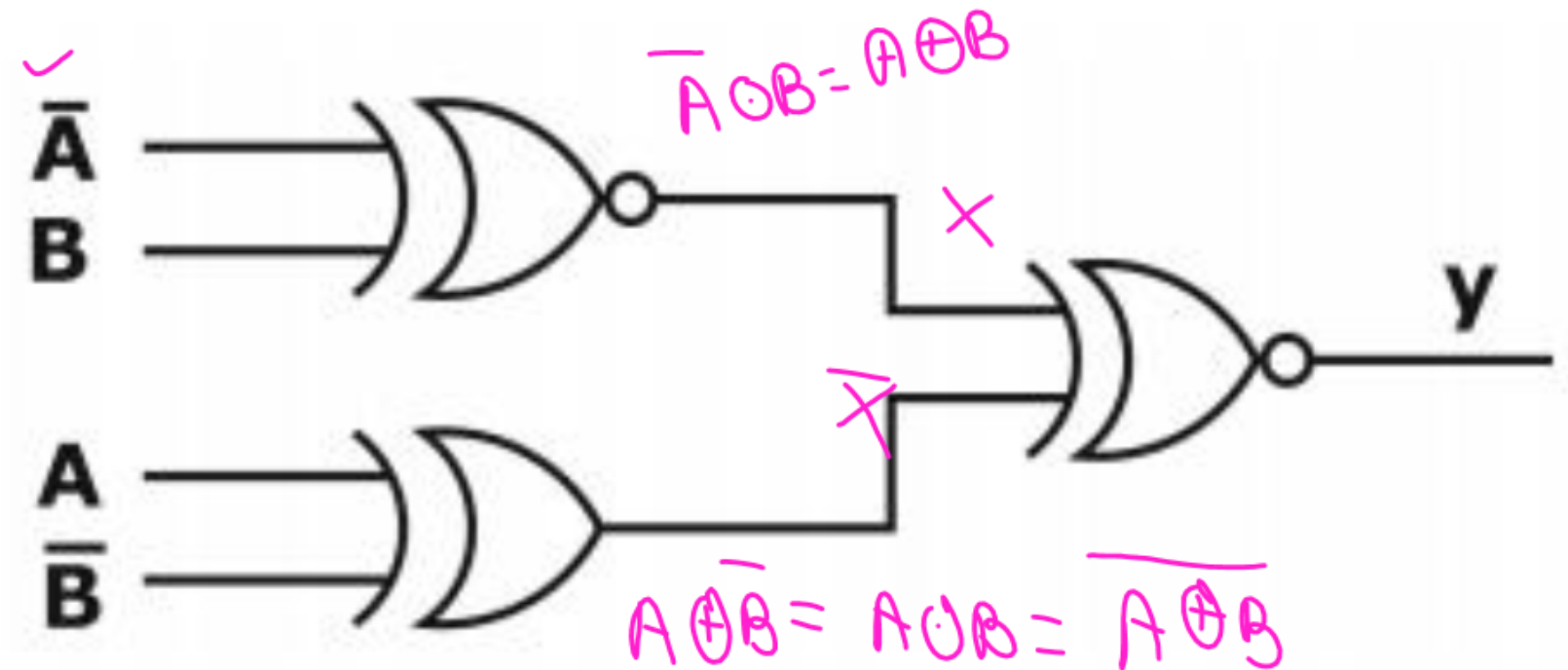
1. Output of the function given below-

A. 0

B. 1

C. $A \oplus B$

D. $A \odot B$



Q Find the minimum no. of NAND GATE required to implement

$$f = A + A\bar{B}C + AB\bar{C} + ABC$$

~~(A) 0~~

(B) 1

(C) 6

(D) 9

$$A(1 + \bar{B}C + B\bar{C} + BC)$$

A

i/p

O/p



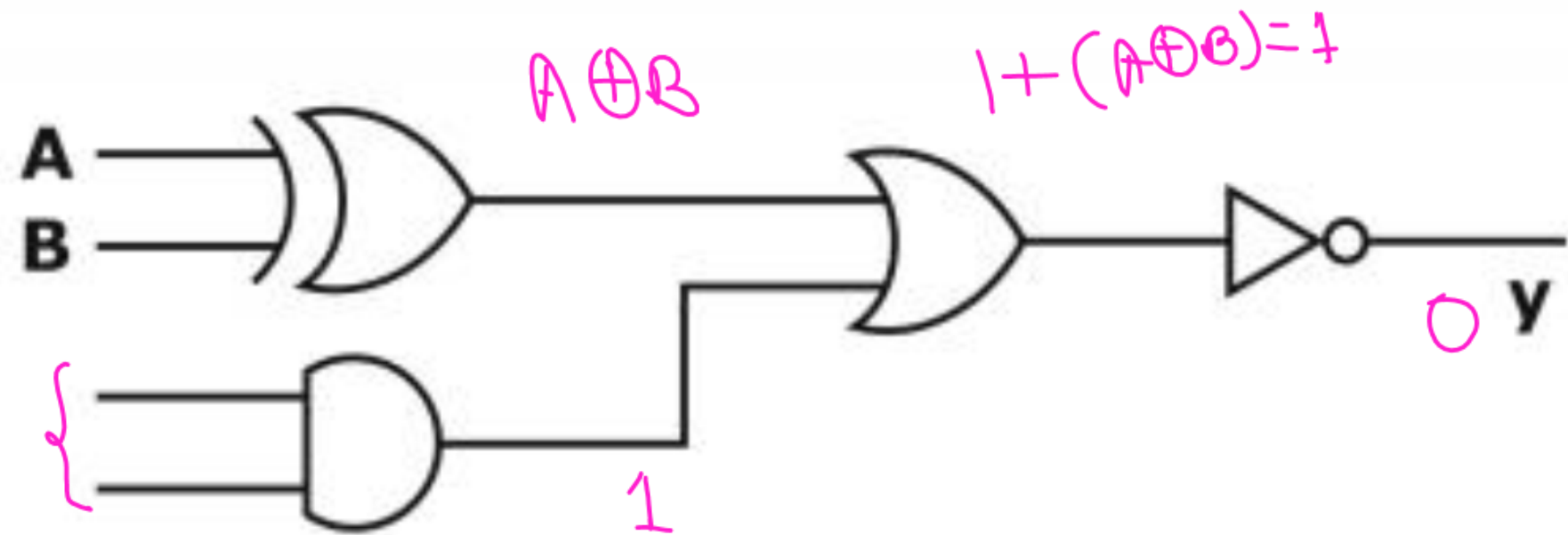
2. The output of the circuit y when all the logics are designed by TTL.

A. 1

B. $A \oplus B$

C. $A \odot B$

☒ D. 0



$$f = \frac{1}{2N \times t_{pd}}$$

$$f = \frac{1}{2 \times 3 \times 2 \times 10^{-9}} \text{ Hz}$$

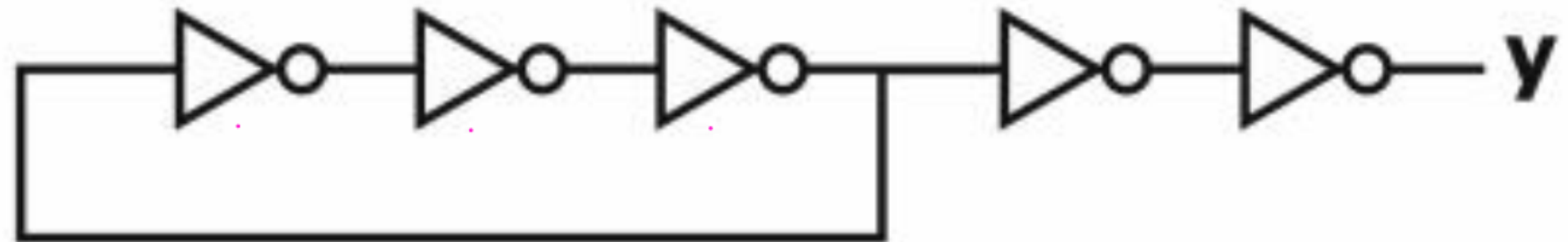
$$f = \frac{1000}{2 \times 3 \times 2} \text{ MHz}$$

$$f = \frac{83.33}{2} \text{ MHz}$$

Astable
Multivibrator

3. The frequency of the signal which is generated by the circuit given below. When all not gates are identical and having propagation delay 2 ns

- A. 56.33 MHz
- B. 43.33 MHz
- C. 0 Hz
- D. None



Boolean Algebra

① Distribution theorems.

$$A + BC = (A + B)(A + C)$$

$$A \cdot (B + C) = A \cdot B + A \cdot C$$

② Consensus theorem.

$$AB + \bar{A}C + BC = AB + \bar{A}C$$

③ Transpose theorem.

$$(A + B)(\bar{A} + C) = AC + \bar{A}B$$

④ D-Morgan's Law

$$\overline{ABC} = \bar{A} + \bar{B} + \bar{C}$$

$$\overline{A + B + C} = \bar{A} \cdot \bar{B} \cdot \bar{C}$$

n. Variable.

total different expression = 2^{2^n}

Self dual expression = $2^{2^{n-1}}$

K-Map (Modified Veitch diagram)

→ Gray code

→ successive no. are differ by one bit

→ unity hamming distance code

→ Cyclic code.

→ Reflecting code.

16 group = $2^{(4)}$ → 4 Variable Reduce.

8 group = $2^{(3)}$ → 3 Variable Reduce.

4 group = $2^{(2)}$ → 2 Variable Reduce.

2 group = $2^{(1)}$ → 1 Variable Reduce.

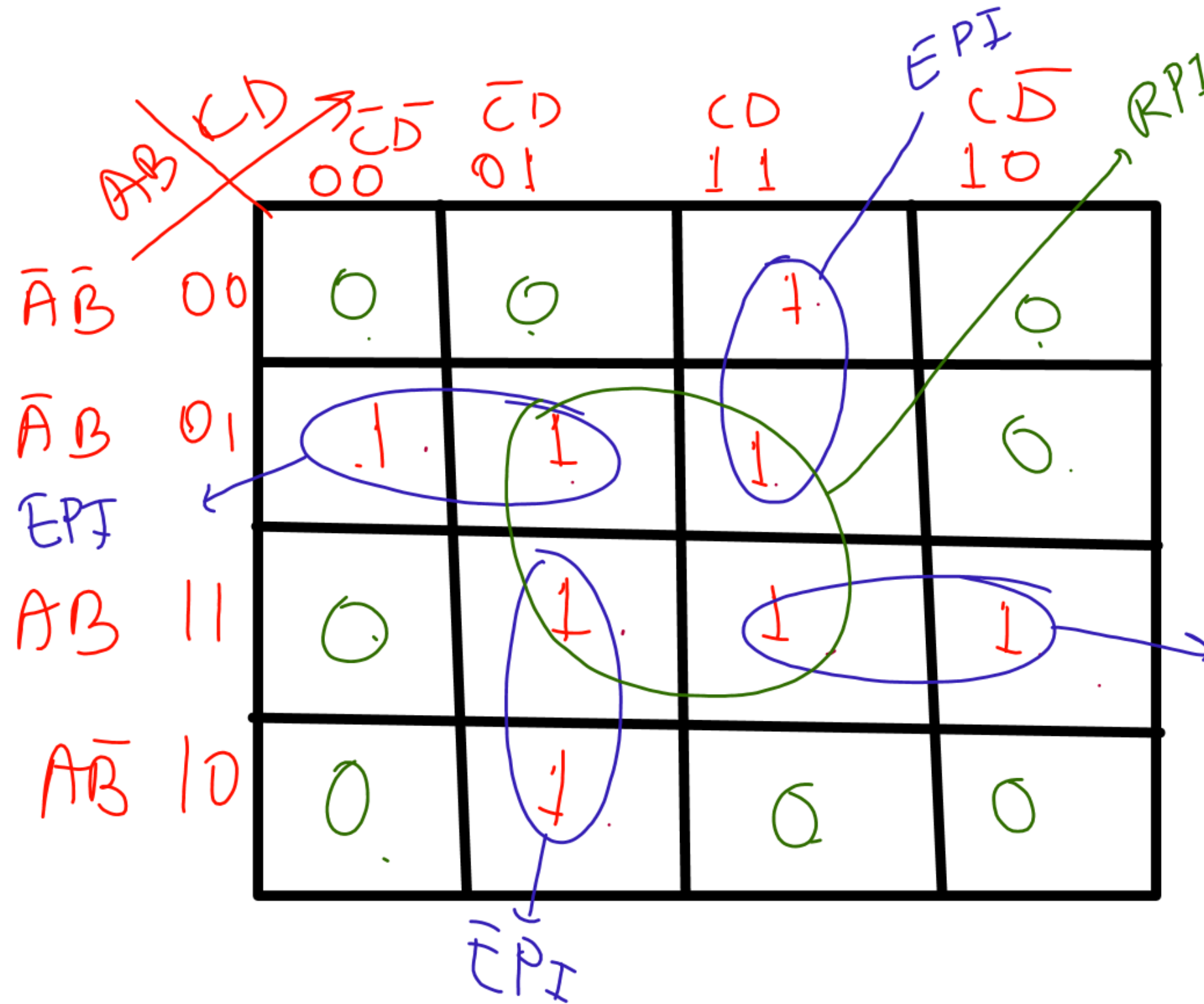
1 group = $2^{(0)}$ → 0 Variable Reduce.

Implicant

Prime Implicant

Essential Prime Implicant

Reduced Prime Implicant



$I = 8$
 $PI = 5$
 $EPI = 4$
 $RPI = 1$

false implicant
 = 8

n variable.

$$\text{Max Prime Implicant} = \underline{\underline{2^{n-1}}}$$

Combinational circuit.

- ① Comparator
- ② MUX
- ③ Decoder
- ④ Parallel adder

✓ Comparator

★ { n Bit comparator

$$\text{Total condition} = 2^{2n}$$

$$\underline{\text{Equal condition}} = 2^n$$

$$\text{unequal condition} = 2^{2n} - 2^n$$

$$\text{Greater} = \text{Less} = \frac{2^{2n} - 2^n}{2}$$

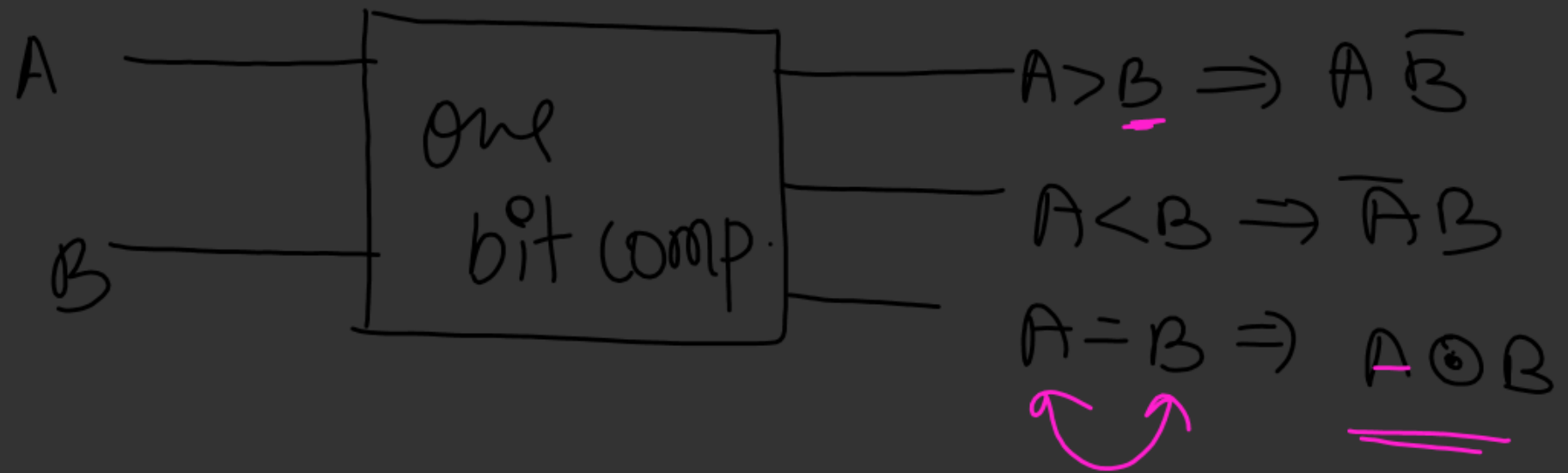
3 Bit

$$T.C = 2^{2 \times 3} = 2^6 = 64$$

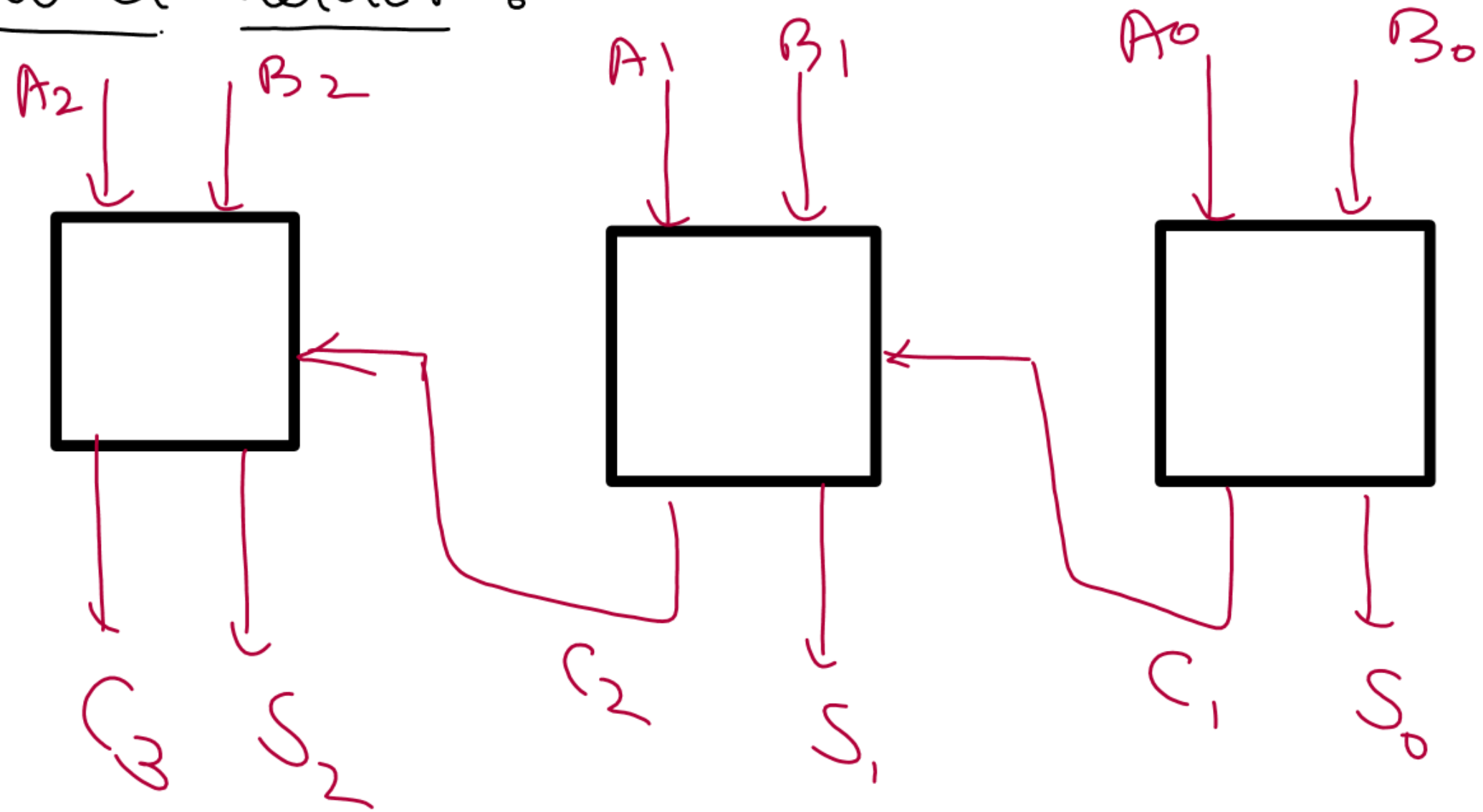
$$E.C = 2^3 = 8$$

$$\text{unequal} = 56$$

$$\text{Greater} = \text{Less} = 28$$



Parallel adder :-



'n' Bit Paralled adder.

- Design
- ① $1 \text{ HA} + (n-1) \text{ FA}$
 - ② $n \cdot \text{FA}$
 - ③ $(2n-1) \text{ HA} + (n-1) \text{ OR}$

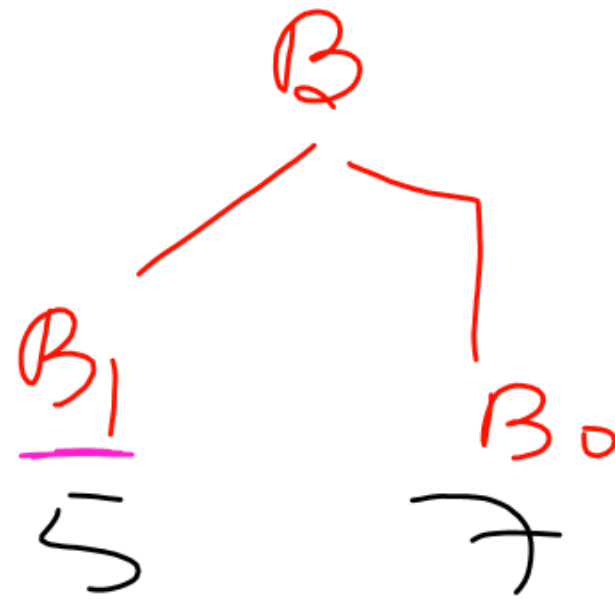
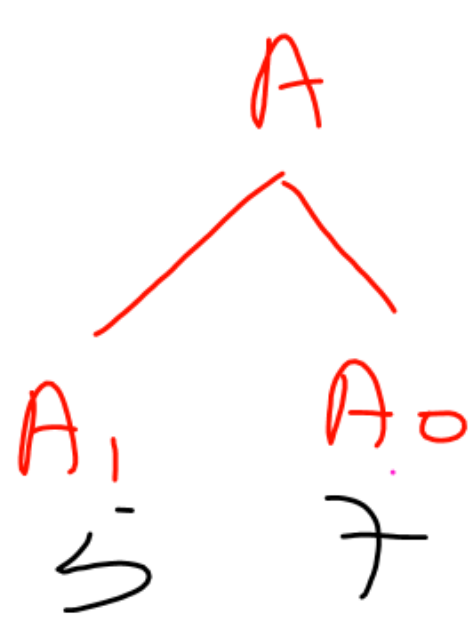
$$1 \text{ FA} = 2 \text{ HA} + 2 \text{ OR}$$

$$\text{Half} \quad \frac{\text{NAND}}{\text{NOR}} \quad \} \quad 5$$

$$\text{Full} \quad \frac{\text{NAND}}{\text{NOR}} \quad \} \quad 9$$

Worst case delay

$$t_{\text{delay}} = (n-1) \tau_{\text{pd carry}} + \text{Max}(\tau_{\text{pd sum}}, \tau_{\text{pd carry}})$$



$$A > B \Rightarrow A_1 \bar{B}_1 + (A_1 0 B_1) \cdot A_0 \bar{B}_0$$

$$A < B \Rightarrow \bar{A}_1 B_1 + (A_1 0 B_1) \cdot \bar{A}_0 B_0$$

$$A = B \Rightarrow (A_1 0 B_1) \cdot (A_0 0 B_0)$$

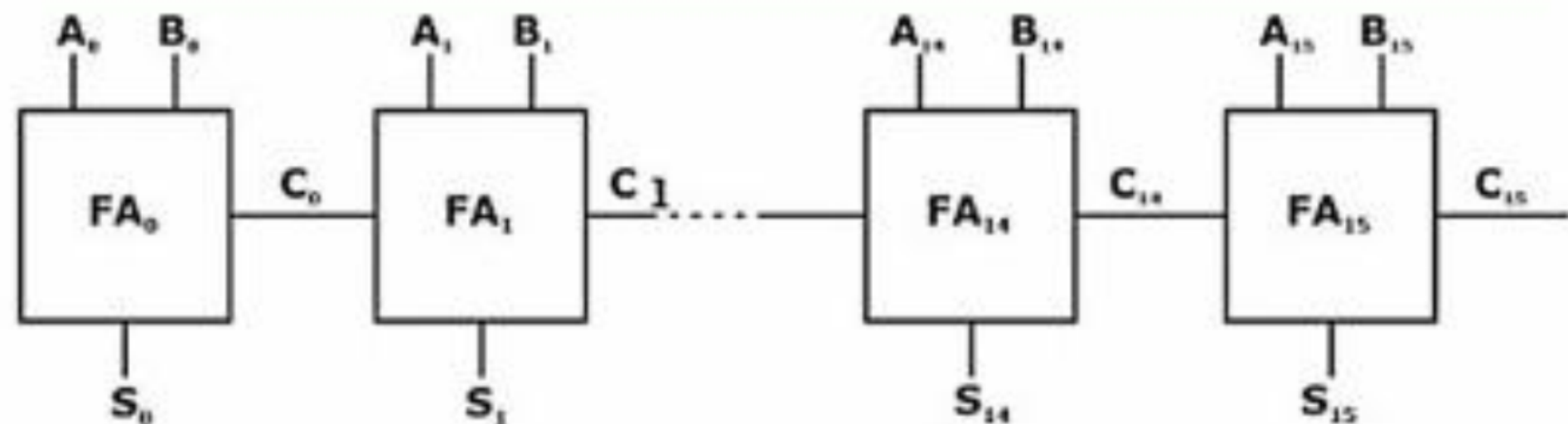
$$(n-1) t_{p_{\text{carry}}} + t_{p_{\text{sum}}}$$

$$15 \times 12 + 15$$

$$= \underline{\underline{195 \text{ ns}}}$$

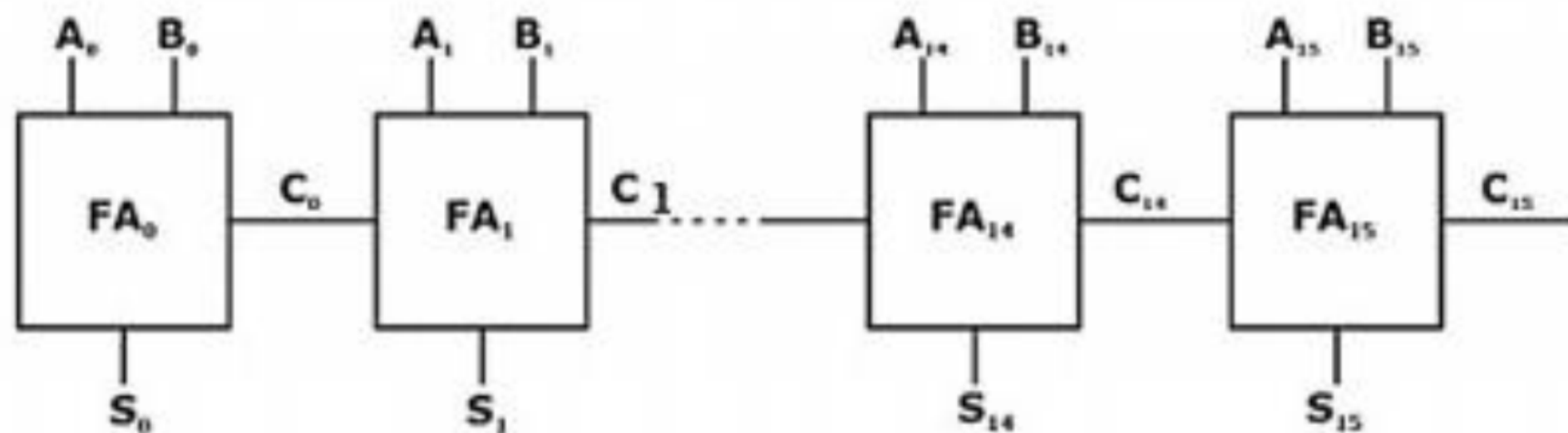
Q. A 16-bit ripple Carry adder is realized using 16 identical full adders (FA) as shown in the figure. The carry-propagation delay of each FA is 12 ns and the sum-propagation delay of each FA is 15 ns. The worst case delay (in ns) of this 16-bit adder will be _____.

- A. 170
- B. 175
- C. 185
- ☒ D. 195



Answer: (D)

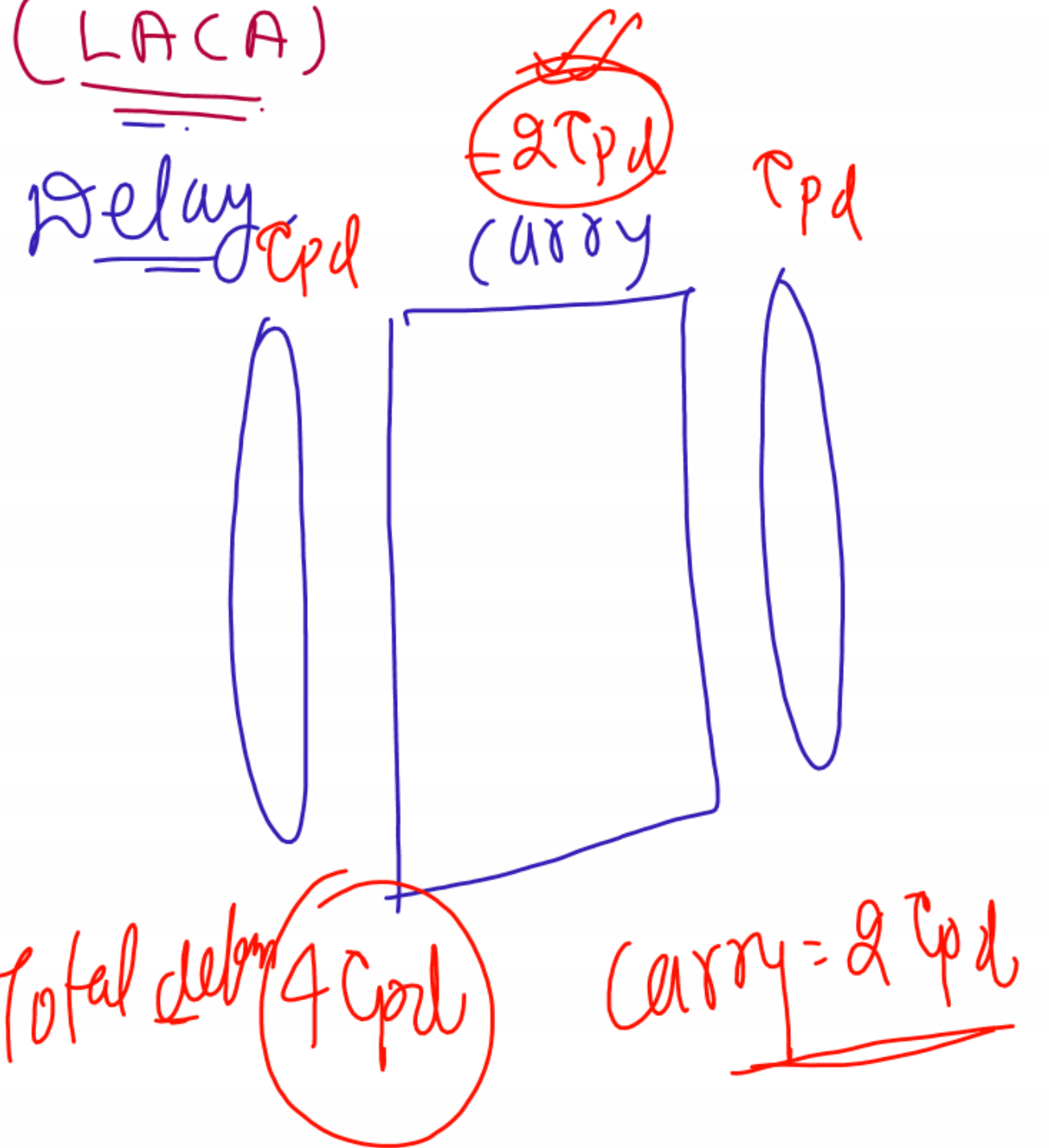
Solution:



Look ahead carry adder (LACA)

n Bit

$$\left\{ \begin{array}{l} \text{no. of AND} = \frac{n(n+1)}{2} \\ \text{No. of OR} = n \end{array} \right\}$$



Q 4 Bit LACA.

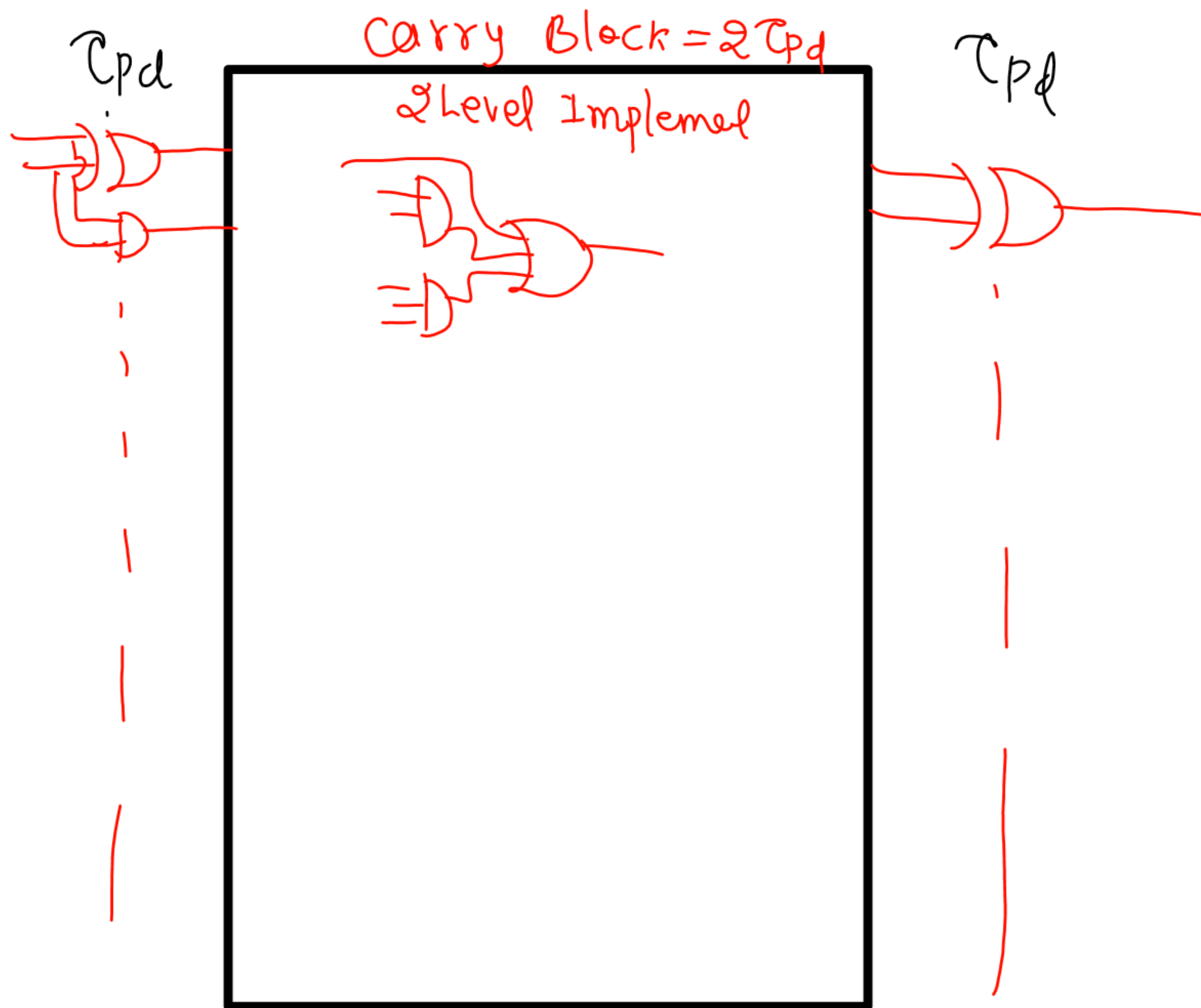
AND, OR = ?

↓
↓
10

↓
4

$$\frac{n(n+1)}{2}$$

n





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